SELVAMURUGAN S

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PROFESSIONAL SUMMARY

Third-year B.Tech Information Technology student with hands-on experience in Python backend development, REST API integration, and AI-powered application development. Familiar with machine learning fundamentals, signal processing, statistical analysis, and anomaly detection. Completed a web development internship and developed projects involving signal analysis, satellite imagery, deepfake detection, and AI image generation. Seeking an AI/ML Engineering Internship to contribute to practical, industry-oriented projects and strengthen machine learning engineering skills.

SKILLS

- **Programming Languages**: Python, JavaScript, Java, SQL, HTML5, CSS3
- **AI & Machine Learning:** Machine Learning Fundamentals, CNN Concepts, Anomaly Detection, Statistical Analysis, Signal Processing
- **Frameworks & Libraries:** Flask, FastAPI, React.js, PyTorch, OpenCV
- **Databases:** MySQL, SQLite, Supabase (PostgreSQL)
- **Tools:** Git, GitHub, VS Code, Postman
- **Core Concepts:** Object-Oriented Programming (OOP), DBMS, REST APIs, Data Structures and Algorithms

EXPERIENCE

Web Developer Intern | Rinex Technologies November 2025 – December 2025

- Developed and tested responsive web application components using HTML5, CSS3, JavaScript, and Flask.
- Integrated frontend interfaces with Flask REST API endpoints to support application data flow.
- Participated in debugging and code reviews to resolve user interface inconsistencies across browsers.

PROJECTS

1. AI-Powered Deepfake Detection System *Python, PyTorch, CNN, OpenCV, Flask, React.js*

- Developed a full-stack application focused on detecting deepfake content across image, video, and audio inputs.
- Worked on media preprocessing, CNN-based deep learning components, and backend API integration.
- Connected the Flask backend with a React.js interface to present analysis results.

2. Signal Analysis Assistant — Smart Signal Intelligence *Python, Signal Processing, Machine Learning*

- Developed a signal analysis workflow for IQ and WAV files with waveform, FFT, and spectrogram visualisations.
- Implemented signal fingerprinting and historical-baseline comparison features.
- Incorporated statistical anomaly detection and Isolation Forest to identify unusual signal patterns.

3. SatQuery AI — Satellite Vision-Language Assistant *Python, AI Application Development, API Integration*

- Developed a project focused on interactive question answering and analysis of satellite imagery.
- Worked on backend application structure and API workflows for AI-assisted image analysis.

4. Gen Image AI — AI Image Generator *Python, Flask, JavaScript, SQLite*

- Developed an AI image-generation web application with user authentication and persistent image history.
- Integrated AI generation APIs through Flask REST endpoints and implemented dynamic frontend rendering.

EDUCATION

Bachelor of Technology, Information Technology Expected 2028
Hindusthan Institute of Technology, Coimbatore
CGPA: 8.05 / 10

CERTIFICATIONS

- Deloitte Australia — Data Analytics Job Simulation (Forage)
- Tata Group — Data Visualisation: Empowering Business with Effective Insights (Forage)
- Fundamentals of Artificial Intelligence — Wadhwani Foundation
- Python Programming Certification
- SQL Certification
- Java Programming and OOP — Saylor Academy
- C Programming — Infosys Springboard